An explicit quartic in 351 variables

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Abstract

We specify an integer polynomial of degree four in 351 total variables whose rational input projection is the integers, under the arithmetic premises of the elliptic source construction. Its complete coefficient table has 4990 nonzero monomials. Three height detectors use sixteen local tests expressed by polynomials whose summands have different valuation classes. Existing formal-logarithm coordinates and nine additional field squares supply the required powers. The construction retains the effective abscissa-sum height bound and the explicit fixed parameter 80989^{54929}. The polynomial has content one and a largest coefficient with 1791357 binary digits.

1 Statement and coefficient specification

The rational input projection of a polynomial $P \in \mathbb{Z}[X, \boldsymbol{Y}]$ is the set of $x \in \mathbb{Q}$ for which $P(x, \boldsymbol{y}) = 0$ has a solution in rational values of $\boldsymbol{Y}$. Variable counts include the input X, and degree always means total degree. The coefficient height $H(P)$ is the largest absolute integer coefficient.

For an integer $N \geq 1$ put $A_N = 80989^{2N+1}$. The exact tables `quadratic-family351.json` and `polynomial-family351.json` specify 323 primitive integer quadratic rows $Q_{N,j}$ in 351 coordinates and

$$P_N = \sum_{j=1}^{323} Q_{N,j}^2.$$

The powers of A_N are recorded separately in the tables solely to abbreviate integer coefficients. The parameter is not a variable.

Theorem 1.1. *Under the arithmetic premises in Section 2, for every integer $N \geq 27464$ and every $x \in \mathbb{Q}$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \boldsymbol{y} \in \mathbb{Q}^{350},\ P_N(x, \boldsymbol{y}) = 0.$$

Every such polynomial has degree exactly four, content one, and 4990 nonzero coordinate monomials. Its coefficient height satisfies

$$H(P_N) < 2^{1520} 80989^{4N+2}.$$

In particular, substituting $A = 80989^{54929}$ specifies a single polynomial P_{27464} with the asserted projection. Its largest coefficient has exactly 1791357 binary digits.

2 Arithmetic premises

Put

$$K = \mathbb{Q}(\rho), \quad \rho^2 = 2, \quad d = 3 - \rho, \quad E : y^2 = x^3 - d^2 x,$$
$$R_0 = (1 + \rho, 2\rho), \qquad P = 2R_0 = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32} \right).$$

We retain the following arithmetic contract, stated with its exact source locations in [3, Section 2].

1. The group and index-height assertions are

$$2E(K) = \mathbb{Z}P, \qquad h(x([n]P)) \geq n^2 - 6 \quad (n \neq 0).$$

Here h is absolute logarithmic Weil height, and σ denotes conjugation on $K/\mathbb{Q}$.

2. For $t = u/v$ in lowest terms, $v > 0$, outside $\{0, 1, 2401, -343/32\}$, let $M_5(t)$ be the product of primes $p > 76176$ with a positive odd valuation on at least one of

$$v, \quad u, \quad u - v, \quad u - 2401v, \quad 32u + 343v.$$

The contact-height input is

$$h(t) \leq \mathcal{H} M_5(t)^{19}, \qquad \mathcal{H} = 2^{110} 210^{1800} \left(\prod_{p \leq 76176} p \right)^{19} + 128 < 2^{2093448}.$$

3. The source pole equations exclude negative odd valuations of their nonzero input b at primes outside $\{2, 3, 17\}$. They admit the specified specialization when $0 < b < 17$, its reduced denominator is a square, b is a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and its valuations there are at least $3, 1, 2$, respectively. Their exact normalized seven-coordinate form is retained in the coefficient table.

4. The simultaneous prime-pattern and conjugate reduction-order input holds with arbitrary fixed odd avoidance modulus, the prescribed real approximation, and the finite congruences simultaneously. The selected primes are 1 modulo $8 \cdot 3 \cdot 17$, with $v_2(d_-) = 6$, $v_2(d_+) = 2$ and $\gcd(d_-(p), d_+(p')) = 4$. Its quantitative weighted form is retained: for fixed distinct coefficients c_j, admissible patterns $p_j = u + c_j w^2$, $N'/2 \leq u \leq N'$, $1 \leq w \leq \tau(N')\sqrt{N'}$, have total weight at least $c_0 (N')^{3/2} \tau(N')$, individual weights $O((\log N')^k)$, and $\tau(N')$ may decrease arbitrarily slowly after the fixed avoidance and congruence conditions are imposed.

These are arithmetic premises, not conclusions of coefficient verification. The effective sum-height theorem used below is proved in [4]; it is not an extra conjectural premise.

3 The local tests and the quadratic system

The four root charts are

$$x = h^2, \qquad j^2 = x^2 - d^2,$$

and give nonzero integer multiples Q_i of P. The squareclass argument establishing this assertion, including its converse and all torsion exclusions, is proved in [2, Section 2]. Write their labels as n, a, t, b and set

$$B = \max(|n|, |a|, |t|, |b|), \qquad S = \sum_{i=1}^{4} x(Q_i) = S_0 + S_1 \rho.$$

The three positive rational square scales are

$$1, \qquad S_0^2, \qquad (S_0 + S_1)^2.$$

They are nonzero because $S_0 > 12$ and $S_0 > \sqrt{2}|S_1|$.

For each detector the contact input is $t_* = a_* \lambda$ for one of these scales. Put $\delta = t_* - 1$,

$$F_*(t_*) = t_*(t_* + 1/2)(t_* + 1)(t_* - 1/2399)(t_* + 32/407),$$

$$R_* = \frac{F_*(t_*)}{a_*^5\delta^6}, \qquad q_m = R_*^{1-2m}\delta^{-2m-2}.$$

These rational formulas are used only when their denominators are nonzero; the actual system consists of the polynomial rows specified in the table. The exceptional value $t_* = 1$ has empty contact radical and is explicitly covered by the local proof in [1].

Write $f(Q) = 1/x(Q)$, extended by zero at the identity. The auxiliary points satisfy $U_0 = H$, $V_0 = H - P$, and $Q_i = U_i - V_i$ as group differences on E. The exact addition rows and identity-inclusive charts of [2] enforce these relations.

Define $A_i = f(U_i)$, $B_i = \sigma f(V_i)$, and

$$C_i = L_i - L_i' \quad (1 \le i \le 4), \qquad I = L_1 - \beta,$$
$$D_H = f(H), \qquad G = L_3^2 - L_2(L_2-2)L_3L_4 + (L_2-2)^2L_4^2 - 1.$$

The alternate ratios L_i' already include conjugation. The nine tests A_i, B_i have order one, C_1, I have order ten, and D_H, C_2, C_3, C_4, G have order forty. For an ordered list of tests, use the factor

$$\mathcal{T}(D_0, \dots, D_{s-1}) = 1 + \sum_{j=0}^{s-1} q_{m(D_j)}^{2^j} D_j^{2^{j+1}},$$

where $m(D_j)$ is its stated order. Thus different required orders can occur within one factor.

The detector uses the product of the factors for these five lists:

$$(D_H, C_1, B_1, B_0), \quad (C_2, I, B_2), \quad (C_3, A_2, B_3),$$
$$(C_4, A_3, B_4), \quad (G, A_4, A_1).$$

Each test occurs exactly once. The additional field powers are

$$C_1^4, I^4, A_2^4, A_3^4, A_4^4, \quad B_1^8, B_0^{16}, \quad A_1^4, A_1^8.$$

In particular the sixteenth power of B_0 is obtained by squaring its existing eighth power, and the eighth power of B_1 by squaring its existing fourth power. All additional field powers use quadratic rows. The full normalization formulas and exact coordinate accounting are in [1].

At a contact prime, a nonzero term in position j has valuation 2^j times an odd integer. These valuations are pairwise distinct and different from zero. The unique-minimum argument and the literal trace-clearing rows give every assigned test order at the same place; this is [1, Proposition 1.2]. The argument also proves the nonvanishing needed for the trace quotient. The rational logarithms, point charts, and scalar equations retain their stated forms.

4 Soundness with the explicit parameter

The input label and the scalar normalizer are

$$\beta = 51X^2 + 2504, \qquad r = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1),$$
$$R_s = \frac{A_N r}{1 + 80989^2 r}, \qquad c_s = R_s + 9792.$$

The retained scalar argument gives, with $c = |a-1|$,

$$c \ge 80989^{L_N}, \qquad L_N = \left\lfloor \frac{187(2N-1)}{8} \right\rfloor + 1, \qquad h(\beta) \le \mathcal{H} c^{19/24}.$$

Its exceptional-point exclusions precede every use of the height detectors and are unaffected by their polynomial factors. The integer certificates in [2] give, for $N \ge 27464$,

$$|a| > (40\mathcal{H})^{10}, \qquad \log(48\mathcal{H}) < |a|/40, \qquad h(\beta) < |a|/40. \tag{1}$$

The effective sum-height theorem yields

$$B^2 \le 2h(S_0) + 2h(S_0 + S_1) + 82.$$

Combining it with the contact-height input for the three scales gives: if a nonzero integer J is divisible by the kth powers of all three contact radicals, then

$$B^2 \le 5\mathcal{H}|J|^{19/k}. \tag{2}$$

The formal logarithm comparison uses only the orders $1, 10, 40$, which the displayed factors supply unchanged. Therefore the fortieth powers of the contact radicals divide

$$\Delta = t^2 - a(a-2)tb + (a-2)^2b^2 - 1.$$

If $n - \beta \neq 0$, their tenth powers divide its nonzero reduced numerator J.

Proof of soundness in Theorem 1.1. A rational zero of P_N makes all its quadratic rows zero. If $\Delta \neq 0$, then $|\Delta| \le 14B^4$, and (2) with $k = 40$ gives

$$B^2 \le 5\mathcal{H}(14B^4)^{19/40} < 20\mathcal{H}B^{19/10}.$$

Thus $B^{1/10} < 20\mathcal{H}$, contradicting $B \ge |a|$ and (1). Hence $\Delta = 0$.

The integral Lucas descent in [3, Section 5] applies to all nonzero integer b and $|a| > 7$, giving $\log B > |a| - 3$. If $n - \beta \neq 0$, its reduced numerator satisfies $|J| \le 2B\exp(h(\beta))$. Equation (2) with $k = 10$ would imply

$$\log B < 10\log(20\mathcal{H}) + 19h(\beta) < 29|a|/40,$$

contrary to the Lucas bound. Thus $n = \beta \in \mathbb{Z}$. Write $X = u/v$ in lowest terms with $v > 0$. Integrality of $51X^2 + 2504$ implies $v^2 \mid 51$. Since 51 is squarefree, $v = 1$ and $X \in \mathbb{Z}$. □

5 Completeness and finite verification

For an integer input, the scalar preparation, Lucas labels, and root charts are those of [2]. They admit all nonzero integer indices. The point and prime choices precede the fresh pole witnesses. Proposition 3.1 of [1] proves their simultaneous extension to the present factors: at each selected contact prime the trace is 1 or 2 modulo $p > 2$; at the fixed finite places the same arbitrary preparation depths give the required squareclasses; and on the fixed compact real choice regions all weighted factors tend uniformly to one as $\lambda \to +\infty$. Thus the pole inputs lie in $(0, 17)$ and meet the unchanged pole-realization hypotheses. Equal height scales simply reuse identical witnesses.

All rational coordinate formulas are defined on these completeness witnesses and fill the 323 rows. This proves the reverse implication in Theorem 1.1. The numerical condition $N \ge 27464$ was used only in the unchanged soundness inequalities (1); completeness introduces no further numerical bound on N.

The exact reproduction commands are

```
python build_quartic_mixed.py
python check_quartic_mixed.py
```

The independent program reconstructs the field normalizations and every modified and retained row, and compares the entire expanded coefficient table. It verifies 323 quadratic rows, 351 coordinates, 5005 symbolic terms before the fixed parameter is collected, and 4990 ordinary monomials afterwards. It checks degree four, content one, the uniform coefficient-height bound, and the exact largest-coefficient bit length at $A = 80989^{54929}$. The arithmetic premises in Section 2 remain mathematical inputs.

References

[1] *Simultaneous valuation conditions and explicit quartics*, 9 November 2026, accompanying proof and independent exact verification. PDF; source: `mixed_valuation_tests_20261109_v11.tex`.

[2] *A quartic in 370 variables with specified integer coefficients*, 8 November 2026, complete quadratic specification and effective height proof. PDF; source: `quartic370_20261108_v10.tex`.

[3] *A forced prime condition and an explicit quartic in 392 variables*, 8 November 2026, arithmetic input contract and scalar/Lucas arguments. PDF; source: `quartic392_20261108_v8.tex`.

[4] *Effective heights of sums of elliptic abscissas*, 8 November 2026, effective four-term sum-height theorem and its application to two rational linear forms. PDF; source: `effective_elliptic_sums_20261108_v10.tex`.

[5] E. Glazer, *An explicit quartic defining the integers in the rationals*, Version 13, manuscript and coefficient data, commit `5bbd74f6a479776a332cbe036809d5dddae63720`, https://github.com/elliotglazer/diophantine-z-q.